

TEACHER'S NAME: _____

STUDENT'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

YEAR 12

HALF YEARLY EXAMINATION

2009

**MATHEMATICS
EXTENSION 2**

*Time allowed - Three hours
(Plus five minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Begin each question on a fresh page.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided.

QUESTION 1 (15 Marks)**Marks**

- (a) If $u = 1 - i$ and $v = -2 + 4i$ are complex numbers, evaluate: 3
- (i) $\text{Im}(u + \bar{v})$
- (ii) $\left| \frac{u + v + 1}{u + v + i} \right|$
- (b) Find the two square roots of $-i$ 3
- (c) Write $\left(\frac{1 + \sqrt{3}i}{1 - \sqrt{3}i} \right)^{10}$ in the form $a + bi$ where a and b are real 3
- (d) On an Argand diagram shade the region represented by the complex number z where $\frac{\pi}{4} \leq \arg z \leq \pi$, $1 \leq \text{Im}(z) \leq 3$ and $|z| \leq 4$ 3
- (e) Describe algebraically and with a diagram the locus of z represented by $z\bar{z} + 10(z + \bar{z}) = 21$. What two purely imaginary values of z satisfy this equation? 3

QUESTION 2 (15 Marks)

- (a) (i) Given $|w| = 2$ and $\arg w = \frac{\pi}{6}$, express $\frac{1}{4}w^4$ in mod-arg form 1
- (ii) Hence, on an Argand diagram, plot the points A, B and C which represent $w, iw, \frac{1}{4}w^4$ respectively. 3
- (b) $OPQR$ is a rhombus. O lies at the origin, P is on the positive real axis and R corresponds to the complex number $1 + \sqrt{3}i$.
- (i) Find the complex number corresponding to the point Q 1
- (ii) If the figure is rotated anticlockwise by 60° about O to form a new rhombus $OP'Q'R'$, show this on an Argand diagram and find the complex number corresponding to the vertex at Q' 2
- (c) If $1, w, w^2$ are the 3 cube roots of unity:
- (i) Show that $1 + w$ is a root of the equation $z^3 - 3z^2 + 3z - 2 = 0$ 1
- (ii) Find the integer root of this equation and the third root in terms of w 2
- (iii) Graph these 3 roots on an Argand diagram and find the centre and radius of the circle on which they lie 2
- (d) Find the roots of the equation $z^5 + 1 = 0$ and prove that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$ 3

QUESTION 3 (15 Marks)**Marks**

- (a) Find the equations of the tangents to the curve $x^2 + y^2 = xy + 3$ at the point where $x = 1$ 4
- (b) Let $P(x) = x^5 - x^4 - x^3 - 7x^2 - 20x - 12$. Given that $P(x) = 0$ has a double root and that $P(2i) = 0$, factorise $P(x)$ into a product of real linear and real quadratic factors 4
- (c) The roots of $x^3 + 6x^2 + 5x - 8 = 0$ are α, β and χ . Derive the monic cubic polynomials whose roots are:
- (i) α^2, β^2 and χ^2 2
- (ii) $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\chi}$ 2
- (d) Two of the roots of the quartic equation $x^4 + px^3 + qx^2 + r = 0$, where $r \neq 0$, add to zero. Prove that $q^2 + p^2r = 0$ 3

QUESTION 4 (15 Marks)

- (a) When a polynomial $P(x)$ is divided by $(x-2)$ and $(x-3)$ the remainders are 4 and 9 respectively. Find the remainder when $P(x)$ is divided by $(x-2)(x-3)$. 3
- (b) $P(x)$ is a real polynomial of degree 3 with roots α, β and χ . If $(1-i)$ is one root and $\alpha\beta + \alpha\chi + \beta\chi = -2$, find:
- (i) $\alpha + \beta + \chi$ 2
- (ii) $\alpha\beta\chi$ 1
- (iii) $P(x)$ 1
- (c)
- (i) Given that $(a-b)^5 = a^5 - 5a^4b + 10a^3b^2 - 10a^2b^3 + 5ab^4 - b^5$, expand and simplify $(z - \frac{1}{z})^5$ 1
- (ii) If $z = \cos \theta + i \sin \theta$ show that $z - \frac{1}{z} = 2i \sin \theta$ and $z^n - \frac{1}{z^n} = 2i \sin n\theta$ 2
- (iii) Hence or otherwise express $\sin^5 \theta$ in the form $a \sin 5\theta + b \sin 3\theta + c \sin \theta$ and then solve completely $16 \sin^5 \theta = \sin 5\theta$ 5

QUESTION 5 (15 Marks)**Marks**

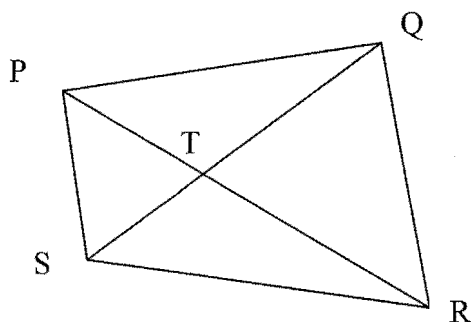
- (a) For the ellipse $x^2 + 4y^2 = 100$ find:
- (i) the eccentricity 1
 - (ii) the coordinates of the focus 1
 - (iii) the equations of the directrices 1
 - (iv) the x and y intercepts 1
 - (v) Neatly sketch the curve showing all the above features 1
- (b)
- (i) Show that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ is given by $ax \sin \theta + by = (a^2 + b^2) \tan \theta$ 3
- (ii) If this normal meets the x-axis at G and if N is the foot of the perpendicular from P to the x-axis, prove that $\frac{OG}{ON} = e^2$ 3
- (c) $P(3p, \frac{3}{p})$ and $Q(3q, \frac{3}{q})$ are 2 points on the rectangular hyperbola $xy=9$ where $p > 0$ and $q > 0$
- (i) Find the equation of the chord PQ 1
- (ii) The tangents at P and Q intersect at M . Find the coordinates of M . 3

QUESTION 6 (15 Marks)

- (a) Show that the curves $x^2 - y^2 = c^2$ and $xy = c^2$ cross at right angles 3
- (b) The straight line $x - 2y + 3 = 0$ cuts the ellipse $x^2 + 2y^2 = 9$ at the points A and B. The tangents to the ellipse at A and B meet at the point C. Find the coordinates of C. 4
- (c) P is the point $(3 \sec \theta, 2 \tan \theta)$ on the hyperbola $4x^2 - 9y^2 = 36$, with focus S and directrix d . The line SP is parallel to an asymptote. Show that the tangent at P and the asymptote meet at a point on the directrix. 4
- (d) PQ is a chord of the rectangular hyperbola $xy = 9$. If PQ has a constant length of 1, show that the locus of the midpoint of PQ has as its equation $4(xy - 9)(x^2 + y^2) - xy = 0$. 4

QUESTION 7 (15 Marks)**Marks**

- (a) PQRS is a cyclic quadrilateral whose diagonals meet at T. A circle is drawn through the vertices of triangle PQT. Prove that the tangent to this circle at T is parallel to SR. 4



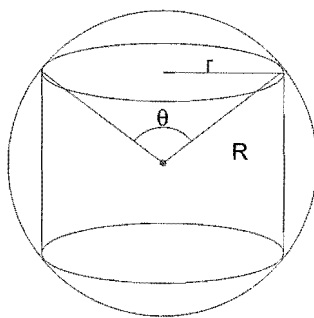
- (b) (i) By considering the expansions of $\cos(A+B)$ and $\cos(A-B)$ show that $\cos(A+B) + \cos(A-B) = 2\cos A \cos B$ 1

(ii) Hence show that $\cos X + \cos Y = 2 \cos \frac{X+Y}{2} \cos \frac{X-Y}{2}$ 1

(iii) Hence solve $\cos \theta + \cos 3\theta = 0$ for $0 \leq \theta \leq 2\pi$. 2

- (c) Solve the inequality $\frac{x-3}{x^2-x} \geq -2$ 4

- (d) A cylinder of height h and radius r is inscribed in a sphere of radius R



(i) Show that $r^2 = \frac{R^2(1 - \cos \theta)}{2}$ 1

(ii) Show that $h^2 = 2R^2(1 + \cos \theta)$ 1

(iii) Hence show that the maximum curved surface area of the cylinder occurs when $h = R\sqrt{2}$ 1

PTO FOR QUESTION 8

QUESTION 8 (15 Marks)

- (a) Using the principle of mathematical induction prove that $a^n + b^n$ is divisible by $a + b$ for all odd positive values of n . 4
- (b) If $y = (x+1)^x$ find $\frac{dy}{dx}$ 2
- (c) Sketch the curve $y = 2 + \frac{1}{x^2 - 1}$, showing the location and nature of all stationary points and the equations of any asymptotes. 4
- (d) Find the area enclosed between the curves $y = \cos x$, $y = \sin x$, and $y = \tan x$ in the domain $0 \leq x \leq \frac{\pi}{2}$. Draw a clear diagram to illustrate the area involved. 5

END OF EXAM

BHHS MATHEMATICS EXT 2
HALF YEARLY 2009
SOLUTIONS

Q1(a)

$$(i) \quad g_m(u+v) = g_m(1-i-2-4i) \\ = g_m(-1-5i) \\ = -5$$

$$(ii) \quad \left| \frac{u+v+1}{u+v+i} \right| = \left| \frac{1-i-2+4i+1}{1-i-2+4i+i} \right| \\ = \left| \frac{3i}{-1+4i} \right| \\ = \frac{|3i|}{|-1+4i|} \\ = \frac{3}{\sqrt{1^2+4^2}} \\ = \frac{3}{\sqrt{17}}$$

b) let $x+iy = \sqrt{-i}$

$$\therefore (x+iy)^2 = -i$$

$$x^2 - y^2 + 2ixy = -i$$

Eq. real & Im parts

$$x^2 - y^2 = 0$$

$$2xy = -1$$

$$y = -\frac{1}{2x}$$

$$\therefore x^2 - \frac{1}{4x^2} = 0$$

$$4x^4 = 1$$

$$x^4 = \frac{1}{4}$$

$$x^2 = \pm \frac{1}{2}$$

But $x^2 \geq 0 \therefore x^2 = \frac{1}{2}$

$$\therefore x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore y = \frac{-1}{\pm \frac{2}{\sqrt{2}}}$$

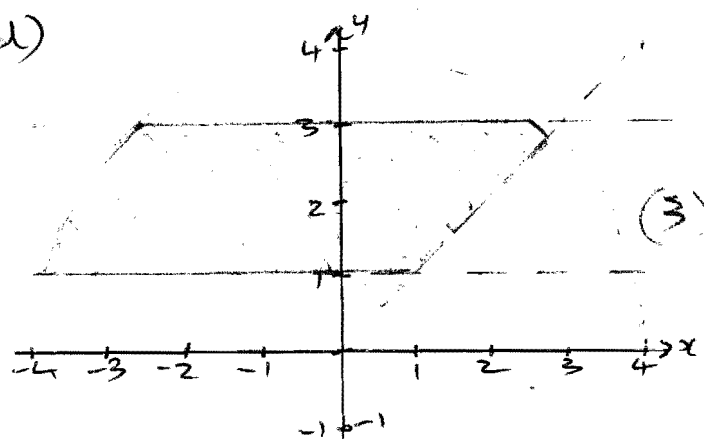
$$= \mp \frac{1}{\sqrt{2}}$$

$$\therefore x+iy = \pm \frac{1}{\sqrt{2}} \mp \frac{i}{\sqrt{2}}$$

$$c) \quad \left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i} \right)^{10} = \left[\frac{2(\frac{1}{2} + \frac{\sqrt{3}}{2}i)}{2(\frac{1}{2} - \frac{\sqrt{3}}{2}i)} \right]^{10} \\ = \frac{2^{10} (\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})^{10}}{2^{10} (\cos \frac{\pi}{3} - i \sin \frac{\pi}{3})^{10}} \\ \text{(by De Moivre)} = \frac{\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3}}{\cos \frac{10\pi}{3} - i \sin \frac{10\pi}{3}}$$

$$= \frac{\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}}{\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}} \\ = \cos \left(\frac{2\pi}{3} - \frac{2\pi}{3} \right) + i \sin \left(\frac{2\pi}{3} - \frac{2\pi}{3} \right) \\ = \cos \frac{0}{3} + i \sin \frac{0}{3} \\ = -\frac{1}{2} + \frac{\sqrt{3}i}{2} \quad -1 \quad (3)$$

d)



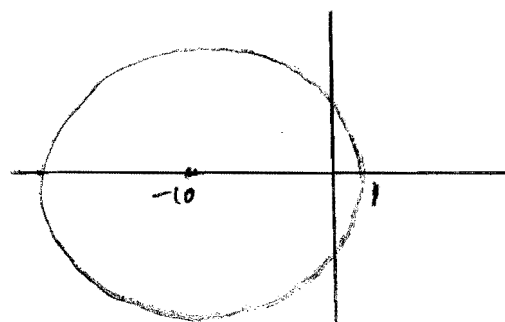
e) $z\bar{z} + 10(z+\bar{z}) = 21$

$$x^2 + y^2 + 20x = 21$$

$$x^2 + 20x + 100 + y^2 = 121$$

$$(x+10)^2 + y^2 = 121 - 100$$

$\therefore$ locus of z is a circle, centre $(-10, 0)$, radius 11



When $x=0$

$$100 + y^2 = 121$$

$$y^2 = 21$$

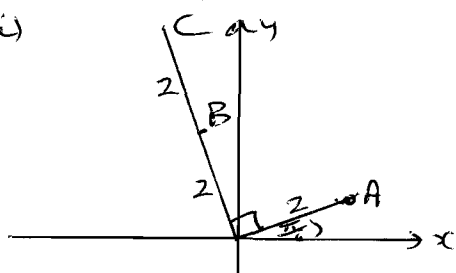
$$y = \pm \sqrt{21}$$

$$\therefore z = \pm \sqrt{21}i \quad -1 \quad (2)$$

Q2(a)(i)

$$\begin{aligned}\frac{1}{4}\omega^4 &= \frac{1}{4} [2(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6})]^4 \\ &= \frac{1}{4} [16(\cos \frac{4\pi}{6} + i\sin \frac{4\pi}{6})] \text{ by De M.} \\ &= 4(\cos \frac{2\pi}{3} + i\sin \frac{2\pi}{3})\end{aligned}$$

(ii)



(4)

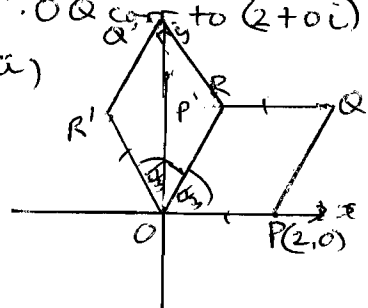
- 3

(b)(i) $|OR| = \sqrt{1+3^2} = 2$

$\therefore OP$ corr. to $2+0i$

$\therefore OQ$ corr. to $(2+0i) + (1+i\sqrt{3}) = 3+i\sqrt{3}$

(ii)



$$Q' = \cos 60^\circ (3+i\sqrt{3})$$

$$= (\frac{1}{2} + i\frac{\sqrt{3}}{2})(3+i\sqrt{3})$$

$$= \frac{3}{2} + i\frac{\sqrt{3}}{2} + \frac{3\sqrt{3}i}{2} - \frac{3}{2}$$

$$= 2\sqrt{3}i$$

(3)

(c)(i) $1+\omega+\omega^2=0 \therefore 1+\omega=-\omega^2$

$\therefore$ Let $z = -\omega^2$

$$LHS = (-\omega^2)^3 - 3(-\omega^2)^2 + 3(-\omega^2) - 2$$

$$= -\omega^6 - 3\omega^4 - 3\omega^2 - 2$$

$$= -(\omega^3)^2 - 3\omega^3 \cdot \omega - 3\omega^2 - 2$$

$$= -1 - 3\omega - 3\omega^2 - 2$$

$$= -3(1+\omega+\omega^2)$$

$$= -3 \times 0 = 0 = RHS.$$

$\therefore 1+\omega$ is a root.

(ii) $P(1) = 1-3+3-2 = -1 \neq 0$

$P(2) = 8-12+6-2 = 0$

$\therefore z=2$ is an integer root

$$\Sigma \alpha = \alpha + 2 + (1+\omega) = -\frac{b}{a} = 3$$

$$\alpha + 3 + \omega = 3$$

$$\alpha = -\omega$$

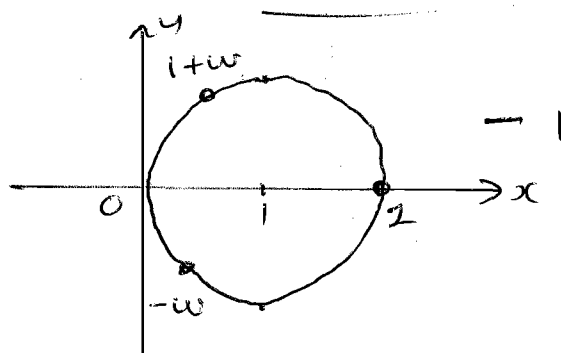
(iii) Roots are: $2 = 1+1$

$$1+\omega = 1+\omega$$

$$-\omega = 1+\omega^2$$

$\therefore$ Points translated 1 unit to the right of those for $1, \omega, \omega^2$

$\therefore$ Circle, Centre $(1,0)$
radius 1



(5)

d) $z^5 = -1$, let $z = \cos \theta$

$\therefore (\cos \theta)^5 = \cos (2k+1)\pi$, $k=0,1,2,3,4$

By De Moivre

$$\cos 5\theta = \cos (2k+1)\pi$$

$$\therefore 5\theta = (2k+1)\pi$$

$$\theta = \frac{(2k+1)\pi}{5}$$

If $k=0$, $z_1 = \cos \frac{\pi}{5} + i\sin \frac{\pi}{5}$

If $k=1$, $z_2 = \cos \frac{3\pi}{5} + i\sin \frac{3\pi}{5}$

If $k=2$, $z_3 = \cos \pi + i\sin \pi = -1$

If $k=3$, $z_4 = \cos \frac{2\pi}{5} + i\sin \frac{2\pi}{5}$

$$= \cos \frac{3\pi}{5} + i\sin \frac{3\pi}{5}$$

$$= \cos \frac{3\pi}{5} - i\sin \frac{3\pi}{5} = \bar{z}_2$$

If $k=4$, $z_5 = \cos \frac{4\pi}{5} + i\sin \frac{4\pi}{5}$

$$= \cos \frac{\pi}{5} + i\sin \frac{\pi}{5}$$

$$= \cos \frac{\pi}{5} - i\sin \frac{\pi}{5} = \bar{z}_1$$

$$\therefore z^5 + 1 = (z - z_3)(z - z_1)(z - \bar{z}_1)(z - z_2)(z - \bar{z}_2)$$

$$\therefore (z+1)(z^4 - z^3 + z^2 - z + 1) = (z+1)(z^2 - 2\cos \frac{\pi}{5}z + 1)(z^2 - 2\cos \frac{3\pi}{5}z + 1)$$

Equating coeffs of z , $z^4z^3+z^2-z+1$

$$-2\cos \frac{\pi}{5} - 2\cos \frac{3\pi}{5} = -1$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2} \quad (3)$$

Q3(a) $x^2 + y^2 = xy + 3$
 Diff. Imp. $2x + 2y y' = y + x \cdot 1 \cdot y'$
 $xy' - 2yy' = 2x - y$
 $y'(x - 2y) = 2x - y$
 $y' = \frac{2x - y}{x - 2y}$ -1

When $x = 1$, $1 + y^2 = y + 3$

$y^2 - y - 2 = 0$

$(y - 2)(y + 1) = 0$

$\therefore y = -1 \text{ or } 2$ -1

When $x = 1$, $m_{\text{Tang}} = \frac{2 - (-1)}{1 - 2} = 1$

$\therefore$ Tangent at $(1, -1)$ is

$y + 1 = 1(x - 1)$

$y = x - 2$ -1

When $x = 1$, $m_{\text{Tang}} = \frac{2 - 2}{1 - 4} = 0$

$\therefore$ Tangent at $(1, 2)$ is

$y = 2$ -1 (4)

(b) Since coeffs are real and $P(2i) = 0$

then $P(-2i) = 0$ and factors are

$(x - 2i)(x + 2i) = x^2 + 4$ -1

Since $P(0) = 0$ has a double root x

then $P'(x) = 0$

$\therefore 5x^4 - 4x^3 - 3x^2 - 14x - 20 = 0$ -1

Try factor of 20.

$P'(1) = 5 - 4 - 3 - 14 - 20 \neq 0$

$P'(-1) = 5 + 4 - 3 + 14 - 20 = 0$

Now $P(-1) = -1 - 1 + 1 - 7 + 20 - 12 = 0$ -1

$\therefore x = -1$ is a double root of $P(x) = 0$

$\therefore (x + 1)^2$ is a factor

$\therefore P(x) = (x + 1)^2 (x^2 + 4) Q(x)$

Since $P(x)$ is monic and constant = 12

then $Q(x) = x - 3$

$\therefore P(x) = (x + 1)^2 (x^2 + 4) (x - 3)$ -1 (4)

c) (i) If $y = x^2$, let $x = \sqrt{y}$ in $P(x) = 0$

$\therefore (\sqrt{y})^3 + 6(\sqrt{y})^2 + 5\sqrt{y} - 8 = 0$ -1

$\sqrt{y}(y + 5) = 8 - 6y$

square both sides

$y(y + 5)^2 = (8 - 6y)^2$

$y(y^2 + 10y + 25) = 64 - 96y + 36y^2$

$y^3 - 26y^2 + 121y - 64 = 0$ -1

or $x^3 - 26x^2 + 121x - 64 = 0$ -1

(ii) If $y = \frac{1}{x}$, let $x = \frac{1}{y}$ in $P(x) = 0$

$(\frac{1}{y})^3 + 6(\frac{1}{y})^2 + 5(\frac{1}{y}) - 8 = 0$ -1

Mult by y^3

$1 + 6y + 5y^2 - 8y^3 = 0$

or $8x^3 - 5x^2 - 6x - 1 = 0$ -1 (4)

i.e. $x^3 - \frac{5}{8}x^2 - \frac{3}{4}x - \frac{1}{8} = 0$

d) let roots be $\alpha, -\alpha, \beta, \gamma$

$\Sigma \alpha = \beta + \gamma = -p$ --- (1)

$\Sigma \alpha \beta = -\alpha^2 + \alpha\beta + \alpha\gamma - \alpha\beta - \alpha\gamma + \beta\gamma = 0$

i.e. $\beta\gamma - \alpha^2 = 0$ --- (2)

$\Sigma \alpha \beta \gamma = -\alpha^2\beta - \alpha^2\gamma + \alpha\beta\gamma - \alpha\beta\gamma = -q$

i.e. $-\alpha^2(\beta + \gamma) = -q$ --- (3)

$\alpha\beta\gamma = -\alpha^2\beta\gamma = r$ --- (4)

From (2) $\beta\gamma = \alpha^2$

in (4) $-\alpha^4 = r$

$\alpha^4 = -r$

$\alpha^2 = \sqrt{-r}$, since $\alpha^2 > 0$

In (3) $-\sqrt{-r}x - p = -q$ -1

square both sides

$-r p^2 = q^2$

$\therefore q^2 + p^2 r = 0$ (3)

Q4 (a) when $P(x)$ is div. by $(x-2)(x-3)$
i.e. $x^2 \leq x+6$, then $\deg P(x) < 2$.

$\therefore$ let $R(x) = ax + b$ —

$\therefore P(x) = (x-2)(x-3)Q(x) + ax + b$

$P(2) = 4 \therefore 4 = 0 + 2a + b \dots (1)$ —

$P(3) = 9 \therefore 9 = 0 + 3a + b \dots (2)$

$(2) - (1) \text{ gives } a = 5$

$\therefore \text{In (1)} \quad 4 = 10 + b$

$\therefore b = -6$

$\therefore R(x) = 5x - 6$ — (3)

(b) If coeffs are real and $x = 1-i$
then $\beta = \bar{\alpha} = 1+i$.

Since $\alpha\beta + \alpha\gamma + \beta\gamma = -2$ —

then $2 + (1-i)\gamma + (1+i)\gamma = -2$

$\gamma - i\gamma + \gamma + i\gamma = -4$

$\gamma = -2$

$\therefore \alpha + \beta + \gamma = (1-i) + (1+i) + -2 = 0$ —

(ii) $\alpha\beta\gamma = (1-i)(1+i)x^{-2} = -4$ —

(iii) $P(x) = x^3 - (\alpha + \beta + \gamma)x^2 - \alpha\beta\gamma = 0$
 $= x^3 - 2x + 4$ — (4)

(c) (i) $z^5 = 5z^4 \frac{1}{2} + 10z^3 \frac{1}{2}z - 10z^2 \frac{1}{2}z^2$
 $+ 5z \frac{1}{2}z^4 - \frac{1}{2}z^5$
 $= z^5 - 5z^3 + 10z - \frac{10}{2} + \frac{5}{2}z^3 - \frac{1}{2}z^5$

(ii) $z - \frac{1}{z} = \cos\theta + i\sin\theta - \frac{1}{\cos\theta + i\sin\theta}$
 $= \cos\theta + i\sin\theta - \frac{\cos\theta - i\sin\theta}{\cos^2\theta + \sin^2\theta}$
 $= 2i\sin\theta$ —

$z^n - \frac{1}{z^n} = z^n - (z^{-n})^{-1} = z^n - z^{-n}$
 $= \cos n\theta + i\sin n\theta - (\cos(n\theta) + i\sin(n\theta))$
 by De Moivre.
 $= \cos n\theta + i\sin n\theta - \cos n\theta - i\sin n\theta$
 $= 2i\sin n\theta$ —

(iii) $(z - \frac{1}{z})^5 = (z^5 - \frac{1}{2}z^5) - 5(z^3 - \frac{1}{2}z^3) + 10(z - \frac{1}{2}z)$
 $\therefore (2i\sin\theta)^5 = 2i\sin 5\theta - 5 \times 2i\sin 3\theta + 10 \times 2i\sin\theta$
 $32i\sin^5\theta = 2i\sin 5\theta - 10i\sin 3\theta + 20i\sin\theta$
 $\therefore \sin^5\theta = \frac{1}{16}\sin 5\theta - \frac{5}{16}\sin 3\theta + \frac{5}{8}\sin\theta$

—

If $16 \sin^5\theta = \sin 5\theta$

then $\sin 5\theta - 5\sin 3\theta + 10\sin\theta = \sin 5\theta$ —

$5\sin 3\theta - 10\sin\theta = 0$

$\sin 3\theta - 2\sin\theta = 0$

$3\sin\theta - 4\sin^3\theta - 2\sin\theta = 0$

$\sin\theta - 4\sin^3\theta = 0$

$\sin\theta(1 - 4\sin^2\theta) = 0$

$\sin\theta = 0$ or $\sin^2\theta = \frac{1}{4}$ —
 $\sin\theta = \pm \frac{1}{2}$

$\therefore \theta = 0, \pm\pi, \pm 2\pi, \dots$

or $\theta = \pm \frac{\pi}{6}, \pm \frac{5\pi}{6}, \pm \frac{7\pi}{6}, \dots$

i.e. $\theta = n\pi, n \in \mathbb{Z}$ —

$\theta = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (8)

Q5

$$(a)(i) x^2 + 4y^2 = 100$$

$$\frac{x^2}{100} + \frac{y^2}{25} = 1$$

$$a = 10, b = 5$$

$$b^2 = a^2(1 - e^2)$$

$$1 - e^2 = \frac{b^2}{a^2}$$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$$= 1 - \frac{25}{100}$$

$$= \frac{75}{100} = \frac{3}{4}$$

$$e = \frac{\sqrt{3}}{2} \quad -1$$

$$(ii) \text{Focus} = (\pm ae, 0) \\ = (\pm 5\sqrt{3}, 0) \quad -1$$

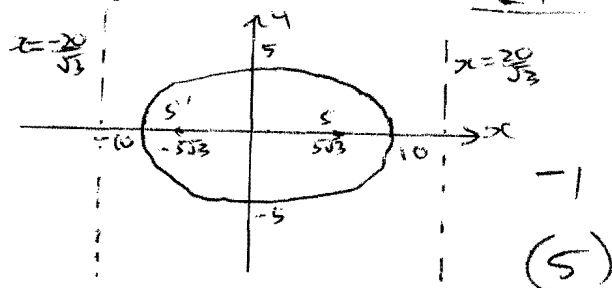
$$(iii) \text{Dir. ax. } x = \pm \frac{a}{e}$$

$$x = \pm \frac{10}{\frac{\sqrt{3}}{2}}$$

$$x = \pm \frac{20}{\sqrt{3}} \quad -1$$

$$(iv) x\text{-axis} = a = 10 \therefore (10, 0) \quad -1$$

$$(v) y\text{-axis} = b = 5 \therefore (0, 5)$$



$$(b)(i) \text{ If } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

diff imp:

$$\frac{2x}{a^2} - \frac{2y}{b^2} \cdot y' = 0$$

$$\frac{yy'}{b^2} = \frac{x}{a^2}$$

$$y' = \frac{b^2 x}{a^2 y} \quad -1$$

$$\text{At P, } m_T = \frac{b^2 a \sec \theta}{a^2 b \tan \theta} \\ = \frac{b}{a \sin \theta} \quad -1$$

$$m_N = -\frac{a \sin \theta}{b} \quad -1$$

∴ The normal is

$$y - b \tan \theta = -\frac{a \sin \theta}{b} (x - a \sec \theta) \quad -1$$

$$by - b^2 \tan \theta = -ax \sin \theta + a^2 \frac{\sin \theta}{\cos \theta}$$

$$ax \sin \theta + by = (a^2 + b^2) \tan \theta \quad (3)$$

$$(ii) \text{At G, } y = 0 \therefore x = \frac{(a^2 + b^2) \tan \theta}{a \sin \theta}$$

$$x_G = \frac{(a^2 + b^2)}{a \cos \theta} \quad -1$$

$$x_N = a \sec \theta$$

$$\therefore \frac{OG}{ON} = \frac{a^2 + b^2}{a \cos \theta} \times \frac{1}{a \sec \theta} \\ = \frac{a^2 + b^2}{a^2} = 1 + \frac{b^2}{a^2} \quad -1$$

$$\text{Since } b^2 = a^2(e^2 - 1) \quad -1$$

$$\frac{b^2}{a^2} = e^2 - 1$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$\therefore \frac{OG}{ON} = e^2 \quad (3)$$

$$d) \frac{y - \frac{3}{p}}{x - 3q} = \frac{\frac{3}{p} - \frac{3}{q}}{\frac{3}{p} - 3q} = \frac{3(q-p)}{\frac{p^2}{p^2 - 3q^2}} \\ = -\frac{1}{pq}$$

$$pqy - 3p = 3q - x$$

$$x + pqy = 3(p+q) \quad -1$$

$$\text{If } xy = q, y = \frac{q}{x} \therefore y' = -\frac{q}{x^2}$$

$$\text{At P}(3p, \frac{3}{p}), m_{\text{Tang}} = -\frac{q}{p^2} = -\frac{1}{p^2}$$

$$\text{At Q}(3q, \frac{3}{q}), m_{\text{Tang}} = -\frac{1}{q^2}$$

$$\therefore \text{Tang at P is } y - \frac{3}{p} = -\frac{1}{p^2}(x - 3p)$$

$$p^2 y - 3p = 3p - x$$

$$x + p^2 y = 6p \quad -1$$

$$\text{and Tang at Q is } x + q^2 y = 6q$$

At point of int.

$$6p - p^2 y = 6q - q^2 y$$

$$y(p^2 - q^2) = 6(p - q)$$

$$y = \frac{6}{p+q} \quad -1$$

$$\therefore x = 6p - p^2 \left(\frac{6}{p+q} \right)$$

$$= 6p - \frac{6p^2}{p+q}$$

$$= \frac{6p^2 + 6pq - 6p^2}{p+q}$$

$$= \frac{6pq}{p+q} \quad -1$$

$$\therefore M = \left(\frac{6pq}{p+q}, \frac{6}{p+q} \right) \quad (4)$$

Q6(a) If $x^2 - y^2 = c^2$ then

$$2x - 2yy' = 0$$

$$y' = \frac{x}{y}$$

If $P(x_1, y_1)$ is the pt of intersection

$$m_{\text{tangent}} = \frac{x_1}{y_1}$$

$$\text{If } xy = c^2 \text{ then } y = \frac{c^2}{x}$$

$$\text{and } y' = -\frac{c^2}{x^2}$$

$$\therefore m_{\text{tangent}} = -\frac{c^2}{x_1^2} = -\frac{x_1 y_1}{x_1^2} = -\frac{y_1}{x_1}$$

$$\therefore m_1 m_2 = \frac{x_1}{y_1} \times -\frac{y_1}{x_1} = -1$$

$\therefore$ curves meet at right angles (3)

b) $x - 2y + 3 = 0$ --- (1)

$x^2 + 2y^2 = 9$ --- (2)

$$(2y - 3)^2 + 2y^2 = 9$$

$$4y^2 - 12y + 9 + 2y^2 = 9$$

$$6y^2 - 12y = 0$$

$$6y(y - 2) = 0$$

$$y = 0, y = 2$$

$$\therefore x = -3, x = 1$$

$$\therefore A = (-3, 0), B = (1, 2)$$

If $x^2 + 2y^2 = 9$

Differentiating $2x + 4yy' = 0$

$$y' = -\frac{x}{2y}$$

At A, $m_{\text{tangent}} = \frac{3}{0} \therefore$ Vertical tangent

$\therefore$ Eqn is $x = -3$ --- (1)

At B, $m_{\text{tangent}} = -\frac{1}{4}$

$\therefore$ Tangent is $y - 2 = -\frac{1}{4}(x - 1)$

$$4y - 8 = -x + 1$$

$$x + 4y - 9 = 0$$

At C, $x = -3 \therefore y = 3$

$\therefore C = (-3, 3)$ --- (4)

(c) $\frac{x^2}{9} - \frac{y^2}{4} = 1 \therefore a = 3, b = 2$

Since $b^2 = a^2(e^2 - 1)$ then $e^2 = 1 + \frac{b^2}{a^2}$

$$e^2 = \frac{13}{9} \therefore e = \frac{\sqrt{13}}{3}$$

$\therefore S, S' = (\pm ae, 0) = (\pm \sqrt{13}, 0)$ --- (1)

Dir are $x = \pm \frac{a}{e} \therefore x = \pm \frac{9}{\sqrt{13}}$

$m_{ps} = m_{\text{asym}}$

$$\therefore \frac{2 \tan \theta - 0}{3 \sec \theta - \sqrt{13}} = -\frac{2}{3}$$

$$6 \tan \theta = -6 \sec \theta + 2\sqrt{13}$$

$$6 \tan \theta + 6 \sec \theta = 2\sqrt{13}$$

$$2 \tan \theta + 2 \sec \theta = \frac{2\sqrt{13}}{3} \quad -1$$

If $4x^2 - 9y^2 = 36$

Differentiating $8x - 18yy' = 0$

$$y' = \frac{4x}{9y}$$

$$\therefore m_{\text{tangent}} = \frac{12 \sec \theta}{18 \tan \theta} = \frac{2 \sec \theta}{3 \tan \theta}$$

$\therefore$ Tangent is

$$y - 2 \tan \theta = \frac{2 \sec \theta}{3 \tan \theta} (x - 3 \sec \theta)$$

At asymptote $y = -\frac{2x}{3}$ --- (1)

$$\therefore -\frac{2x}{3} - 2 \tan \theta = \frac{2 \sec \theta}{3 \tan \theta} (x - 3 \sec \theta)$$

$$-2x \tan \theta - 6 \tan^2 \theta = 2x \sec \theta - 6 \sec^2 \theta$$

$$x(2 \sec \theta + 2 \tan \theta) = 6(\sec^2 \theta - \tan^2 \theta)$$

From above

$$\frac{2\sqrt{13}}{3} x = 6 \quad -1 \quad (4)$$

$$x = \frac{9}{\sqrt{13}} = \frac{a}{e} = \text{dir.}$$

(d) If $PQ = 1$ then $(x_2 - x_1)^2 + (y_2 - y_1)^2 = 1$

$$M_{PQ} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{x_1 + x_2}{2}, \frac{\frac{9}{x_1} + \frac{9}{x_2}}{2} \right) = \left(\frac{x_1 + x_2}{2}, \frac{9(x_2 + x_1)}{2x_1 x_2} \right)$$

If $M = (X, Y)$ then $X = \frac{x_1 + x_2}{2}, Y = \frac{9x}{x_1 x_2}$

$$\therefore x_1 x_2 = \frac{9X}{Y}$$

Now if $(x_2 - x_1)^2 + (y_2 - y_1)^2 = 1$ --- (1)

then $(x_2 + x_1)^2 - 4x_1 x_2 + (y_2 + y_1)^2 - 4y_1 y_2 = 1$

$$(2X)^2 - 4x_1 x_2 + (2Y)^2 - 4 \times \frac{9}{x_1} \times \frac{9}{x_2} = 1$$

$$4X^2 - \frac{36X}{Y} + 4Y^2 - \frac{324 \times \frac{Y}{9X}}{1} = 1$$

$$4X^2 - \frac{36X}{Y} + 4Y^2 - \frac{36Y}{X} = 1$$

$$4X^3 Y - 36X^2 + 4X Y^3 - 36Y^2 = XY$$

$$4(X^3 Y - 9X^2 + X Y^3 - 9Y^2) - XY = 0$$

$$4[X^2(XY - 9) + Y^2(XY - 9)] - XY = 0$$

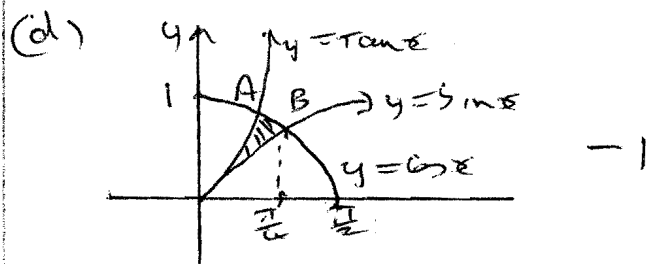
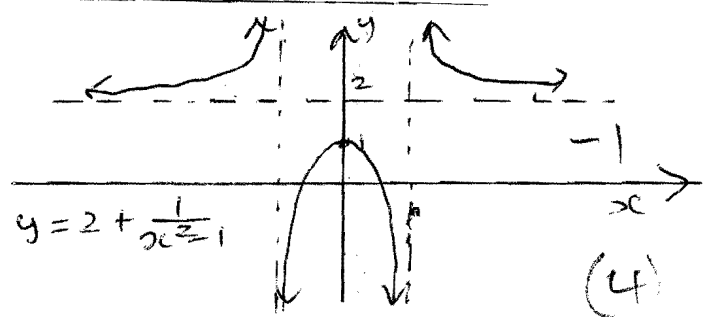
$$4(XY - 9)(X^2 + Y^2) - XY = 0 \quad (4)$$

Q8(a) Prove true for $n=1$
 $a^2 + b^2 = (a+b) \cdot 1$ True for $n=1$
 Assume true for $n=2k-1$
 i.e. $a^{2k-1} + b^{2k-1} = (a+b)P$ — |
 Prove true for $n=2k+1$
 i.e. $a^{2k+1} + b^{2k+1} = (a+b)Q$.
 $LHS = a^2 \cdot a^{2k-1} + b^{2k+1}$
 $= a^2[(a+b)P - b^{2k-1}] + b^{2k+1}$
 $= (a+b)a^2P - a^2b^{2k-1} + b^{2k+1}$
 $= (a+b)a^2P - b^{2k-1}(a^2 - b^2)$
 $= (a+b)a^2P - b^{2k-1}(a-b)(a+b)$
 $= (a+b)[a^2P - (a-b)b^{2k-1}]$ — |
 $= (a+b)Q$ where
 $Q = a^2P - (a-b)b^{2k-1}$
 $\therefore$ True for $n=2k+1$ if true for $n=2k-1$

Since true for $n=1$ and since true
 for $n=2k+1$ if true for $n=2k-1$
 then it is true for $n=3, 5, 7, \dots$
 $\therefore$ True for all positive odd
 integer values of n . (4)

(b) $y = (x+1)^x$
 $\therefore \ln y = \ln(x+1)^x$
 $= x \ln(x+1)$ — |
 Diff imp.
 $\frac{1}{y} y' = \ln(x+1) \cdot 1 + x \cdot \frac{1}{x+1}$
 $= \ln(x+1) + \frac{x}{x+1}$
 $\therefore y' = \left(\ln(x+1) + \frac{x}{x+1} \right) (x+1)^x$ (2)

(c) $y = 2 + \frac{1}{x^2-1}$
 Vert. Asym. when $x = \pm 1$ — |
 Hor. Asym. when $y = 2$ — |
 $y' = \frac{-2x}{(x^2-1)^2}$
 St. pts when $y' = 0 \therefore x = 0$
 If $x < 0$, $y' > 0$
 If $x > 0$, $y' < 0$ $\therefore$ Max at $(0, 1)$ — |
 $f(x) = f(-x) = 2 + \frac{1}{x^2-1}$ $\therefore$ even.



At A, $\cos x = \tan x$
 $= \frac{\sin x}{\cos x}$
 $\therefore \cos^2 x = \sin x$
 $1 - \sin^2 x = \sin x$
 $\sin^2 x + \sin x - 1 = 0$
 $\sin x = \frac{-1 \pm \sqrt{1+4}}{2} = \frac{-1 \pm \sqrt{5}}{2}$
 Since x is acute $\sin x = \frac{-1 + \sqrt{5}}{2}$ — |

 $y = \sqrt{4 - (-1/2)^2}$
 $= \sqrt{4 - 1/4}$
 $= \sqrt{15/4}$
 $\therefore \cos x = \frac{\sqrt{15}}{2}$ — |

At B, $\sin x = \cos x$
 $x = \frac{\pi}{4}$
 Area = $\int_0^A \tan x dx + \int_A^B \cos x dx - \int_0^B \sin x dx$ — |
 $= [-\ln(\cos x)]_0^A + [\sin x]_A^B + [\cos x]_0^B$
 $= -\ln\left(\frac{\sqrt{5}-1}{2}\right) - \ln 1 + \sin\left(\frac{\pi}{4}\right) - \left(-\frac{1+\sqrt{5}}{2}\right)$
 $+ \cos\left(\frac{\pi}{4}\right) - 1$
 $= -\ln\left(\frac{\sqrt{5}-1}{2}\right) + \frac{1}{\sqrt{2}} + \frac{1+\sqrt{5}}{2} + \frac{1}{\sqrt{2}} - 1$
 $= \sqrt{2} - 1 + \frac{1}{\sqrt{2}} + \frac{1+\sqrt{5}}{2} - \ln\left(\frac{\sqrt{5}-1}{2}\right)$
 $= \sqrt{2} - \frac{1}{2} - \frac{\sqrt{5}}{2} - \ln\left(\frac{\sqrt{5}-1}{2}\right)$ (5)